

SHIVADHARSHAN SHAN LINGESWARAN

AI & Edge Computing Engineer | Autonomous Systems & Firmware Specialist

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"Scientists investigate that which already is; Engineers create that which has never been."

EDUCATION

Imperial College London

2019 - 2020

MRes Molecular Science & Engineering

Miniature Surface Acoustic Wave Sensors for Remote Temperature Measurement in Instrumentation

University of Hertfordshire

2015 - 2018

BSc Physics

Propulsion Systems for Asteroid Mining

Hatch End Sixth Form

A Levels

Mathematics, Physics, Chemistry

TECHNICAL SKILLS

Primary - Daily Use/Production:

Python, C, C#, C++, Bash

Secondary - Regular Project Use:

JavaScript, SQL

Applied / Systems-Level:

Java, Assembly (ARM/x86)

Embedded & Security

Automotive Testing (Vehicle-level & HIL), Reverse Engineering (Ghidra, IDA Pro), CAN Bus, OBD-II, JTAG, SWD, PCB Design, ARM Cortex-M, ESP32, Embedded Linux, Firmware Development & Security Audits.

AI & Computer Vision

TensorFlow, PyTorch, Hugging Face, Ollama, LangChain, OpenCV, YOLO, ANPR, Computer Vision Pipelines, Object Detection, Image Segmentation, Video Analytics, Edge AI, Model Deployment.

Automation & Scripting

Selenium, PyTest, Web Scraping, Task Scheduling, Custom Scripting (Python, Bash), Data Pipelines, Test Automation, System Monitoring.

CERTIFICATIONS

From Linktree Portfolio

🔥 Applied Ethical Hacking & Security

🔥 Automotive Cybersecurity

🔥 Embedded C & AUTOSAR Overview

🔥 Java

🔥 JavaScript

🔥 Python

🔥 C / C++ / C#

PROFESSIONAL PROFILE

Physics-trained engineer specialising in the interface between **AI systems and physical hardware**. I design, analyse, and deploy intelligent systems that operate under real-world constraints—spanning **embedded firmware, automotive ECUs, computer vision, and agent-based AI**. My work combines hands-on reverse engineering (CAN Bus, JTAG, ECU firmware) with higher-level software systems (Python, ML pipelines, cloud deployment), ensuring reliability, safety, and performance from silicon to system level. I bring a rigorous, evidence-led engineering approach, with all technical claims grounded in practical delivery across automotive, mixed reality, and autonomous systems.

WORK EXPERIENCE

HP Tuners

Oct 2023 - Feb 2025

Vehicle Engineer

Hybrid

- **Firmware Reverse Engineering & Tooling Enablement:** Analysed production ECU firmware using Ghidra, IDA Pro, and ARM assembly, examining raw binaries to understand memory layouts, control logic, and calibration structures in support of safe and repeatable vehicle tuning.
- **Internal Analysis Tooling (Python Hex Plotter):** Developed a reusable Python-based hex plotting and visualisation tool to analyse and compare multiple firmware variants without reliance on expensive commercial tooling, providing colour-coded hex-resolution views, plots, and exportable outputs for tuning and validation workflows.
- **Automation of Firmware Workflows:** Consolidated and automated previously standalone internal scripts into a sequenced, repeatable workflow, reducing reliance on tribal knowledge, improving consistency across per firmware analysis runs, and lowering human error during routine firmware preparation.
- **Automotive Testing & Validation:** Supported vehicle-level testing and Hardware-in-the-Loop (HIL) validation, reproducing real-world driving scenarios to identify firmware edge cases ahead of release.
- **Firmware Operations & Release Support:** Performed routine but critical firmware tasks including patching, calibration updates, and validation checks, contributing incremental improvements that fed into release readiness and reduced regression risk.

Syngenta

Technical Specialist

Mar 2022 - Mar 2023

Hybrid

- **Mixed Reality Development (HoloLens 2):** Designed and delivered in-house Mixed Reality systems using Unity and C# to support operational laboratory workflows, enabling hands-free interaction under PPE constraints and reducing friction in safety-critical, regulated environments.
- **QR Code-Based Chemical Tracking:** Implemented dynamic QR code recognition on HoloLens 2 with cloud-based lookups against internal databases, enabling real-time identification and traceability of chemical containers across varied lab conditions.
- **System Accuracy & API Integration:** Integrated the PubChem API into the MR application to support voice-driven chemical identification, resolving spoken input into confidence-ranked candidate compounds and significantly reducing manual keyboard entry and error rates.
- **3D Chemical Visualisation:** Implemented an interactive 3D molecular viewer using structures sourced from PubChem, allowing spatial exploration of chemical compounds within the MR environment to support comprehension and validation workflows.

THOUGHT LEADERSHIP

Technical Articles on Medium

- "Can Any CAD Model Be Converted to Python?"
- "Converting Natural Language Prompts to 3D CAD Objects"
- "Unveiling the Magic of Benford's Law"
- "A Journey into the Mind's Eye with Cutting-Edge Technology"

GITHUB PROJECTS

Exciting Projects To Watch Out For

- ContinuousLearningTradingAgent
- Immersive-ARMRVR
- IOT-Eye
- ArbitrageBotsPionexClone
- DiscoverSpeaker
- NanoBananaBatch
- GoogleCloudCheatSheet

LANGUAGES

English	•••••
Tamil	•••••
Russian	••
Ukrainian	••
Hindi	••
German	••
French	••
Spanish	••
Mandarin	•

Independent R&D Engineer
Consultant (AI & Signal Processing)

Oct 2020 – Present
Hybrid

- Agentic AI Development:** Developed Python-based autonomous systems, including a financial trading agent utilising WebSockets to execute grid strategies with real-time PnL tracking.
- 3D Vehicle Model Generation:** Built pipelines for generating high-resolution 3D vehicle models using photogrammetry and AI-assisted Bill of Materials (BOM) extraction to support virtual analysis and customisation.
- Analytical Tooling & Automation:** Created analytical tools for property investment decision-making, integrating financial models, valuation data, and operational inputs.
- Mixed Reality Prototyping:** Prototyped MR concepts for automotive applications, extending prior MR experience into vehicle-focused visualisation and interaction scenarios.

KEY TECHNICAL PROJECTS

Real-Time Grid Trading Bot FastAPI, React, WebSockets

- Built a dynamic grid trading system with webhook execution, PnL tracking, and market visualisation using TradingView + OANDA API.

Flying Car Prototype eVTOL, CAD Modeling, Advanced Propulsion

- Developed conceptual designs for a modular flying car prototype, integrating cyclorotor technology and innovative propulsion systems for vertical takeoff and landing (VTOL).
- Built detailed CAD models of eVTOL components for structural analysis and optimisation of flight dynamics.

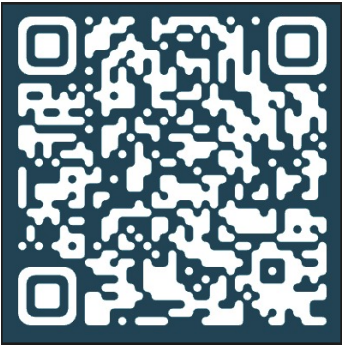
Rover Prototype Development ROS, Embedded Systems, Computer Vision

- Designed and developed a rover for autonomous navigation and object detection using ROS, integrating LIDAR and camera systems for obstacle avoidance and path planning.
- Implemented edge-computing systems for real-time data processing and decision-making on the rover.

PI Stack Edge Computing Raspberry Pi, Docker, Kubernetes

- Built an edge-computing stack using Raspberry Pi clusters, leveraging Docker and Kubernetes for scalable deployment of AI and data processing models.
- Deployed distributed workloads for real-time data analysis, integrating sensors and IoT devices for smart automation systems.

INDEPENDENT TECHNICAL FOCUS



I focus on building end-to-end engineering systems spanning embedded hardware, software, and applied AI. My independent work emphasises automation, real-time data processing, and human-in-the-loop interfaces, with particular interest in edge computing, autonomous systems, and tooling that reduces operational friction in complex technical environments.

Firmware & Tooling	██████████
System Validation	██████████
Automation	██████████
MR Interfaces	██████████
Data Pipelines & APIs	██████████
Edge Computing	██████████
Embedded Systems	██████████
Binary Analysis	██████████
Automotive Systems	██████████
AI Integration	██████████
Computer Vision	██████████
System Reliability	██████████
Human-Machine Interfaces	██████████
Technical Tooling Design	██████████

"A good man leaves an inheritance to his children's children."